

Group Identity and Agency Frictions: Evidence using Big Data*

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Abstract

I examine whether similarity in social identities between a manager and the board affects executive compensation, firm value, and hence, agency frictions. By using a novel dataset on surnames with multiple identities (native language, native place, and caste), developed by merging micro census data of 474 million Indians with Linguistic Survey of India data, I provide evidence that the firms with a shared group identity between a manager and the board do well compare to other firms and due to in-group favoritism, managers of such firms earn higher compensation. These results are stronger for group identity based on native language and native place. I find that the firm benefits from taking on the cost of in-group favoritism as it reduces the agency frictions and increases firm value in the long run. These results are robust to the endogeneity test, influence of a manager in the firm, college ties, ties from past employment, and various other checks.

Keywords: Group Identity, In-group Favoritism, Managerial Compensation, Firm Value

JEL Codes: C92, M12, D21, D22

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1 Introduction

Standard neo-classical models in finance posit that corporate decisions are made virtually in a vacuum and independently of the social identities of agents. But since the seminal work of George A. Akerlof and Rachel E. Kranton (2000), there has been increased interest in social identity research in economics, yielding new insights into phenomena which standard economic analysis on individual-level incentives proves unable to explain. Social identity models have been applied to the analyses of gender discrimination (Kranton and Hall, 2001), the economics of poverty and social exclusion, the household division of labor (Akerlof and Kranton, 2000), contract theory (Akerlof and Kranton, 2005), economic development (Basu, 2006), and public goods provision (e.g., Croson, Marks, and Snyder, 2008; Eckel and Grossman, 2005), summarized in Akerlof and Kranton (2010). Inspired by these works on social psychology of identity and by gaining insight from research on group identity in social preferences (Chen and Li, 2009; Chen and Chen, 2011), this paper examines how shared group identity between a manager and the board affects the executive compensation, agency frictions, and the long term firm value.

Psychologists have long posited a self or “ego” as a primary force of individual behavior. They have further associated an individual’s sense of self to social categories; and individuals identify with people in some categories and differentiate themselves from those in others (Akerlof and Kranton, 2010). Fifty years of research into anthropology, social psychology, and evolutionary biology supports the idea of social categorization, prejudice, and inter-group favoritism (Allport, 1954; Brown, 1988; Devine, 1989; Brewer, 2002); neuroscience also confirms this notion that our brain responds differently to the in-group and out-group members (Amodio, 2008; Ito and Bartholow, 2009; Chiao and Mathur, 2010; Eres and Molenberghs, 2013). Nevertheless, it happens because ‘personal identity’ and ‘social identity’ are governed by the same part of the brain (Volz et al., 2009). A human can assimilate with a

group based on any number of different social traits - race, gender, age, profession, ethnicity, status, country of birth, sports team, social group, political ideology, and education are just a few examples. Research has even shown that people categorize themselves and others based on trivial criteria (Tajfel et al., 1971) and this categorization can be very fluid and is often context dependent (Turner et al., 1994). This form of in-group favoritism is likely to be the primary driver of social biases, such as prejudice, stereotyping, and discrimination (Crisp and Hewstone, 2001).

The ‘Social Identity Perspective’, the leading theory of identity and intergroup relations, combines Social Identity Theory (SIT, Tajfel and Turner, 1979) and Self Categorization Theory (SCT, Turner *et al.*, 1987; Hogg and Abrams 1998; Shih, Pittinsky, and Ambady, 1999). According to SCT, identification with a group is a form of self-categorization in society. An individual belongs to many groups: gender, religious group, nation, race, etc, but he does not identify with all of them at the same time. As shown by several experiments (e.g., Hogg et al. 1998; Shih, Pittinsky, and Ambady, 1999), identification and behavior depends on which groups are salient. For instance, Steele and Aronson (1995) find that African-American students, who are stereotyped to be poor students, underperformed relative to white students when they are told that the test is diagnostic of their abilities. In a similar kind of experiment, Levy (1996) find that elderly people perform worse on a memory task if they have previously been primed with a negative stereotype of the elderly than if they have been primed with a positive stereotype of the elderly. According to social psychology literature, if identity remains in the sub-conscious mind due to strong priming effect and legitimized by groups of people in society and then that identity can give a birth to a stereotype related to that identity. Using gender as an identity, Carlana (2019) shows that implicit stereotypes in teachers regarding gender induce girls to underperform in math and self-select into less demanding high-schools, following recommendation of their teachers.

The recent literature in economics and finance have shown that how in-group favoritism

affects the decision making of loan officers (Fisman, Paravisini, and Vig, 2017), sell-side analysts (Jannati *et al.* 2019), board of governors in M&A deals (Bhalla et al. 2018), employers (Becker, 1957; Bertrand and Mullainathan, 2004; Banerjee et al. 2009, Cornell and Ivo, 1996; Charles Sarsons, 2019), retail investors (Morse and Shive, 2011; Grinblatt and Kelohraju, 2001), and mutual fund investors (Kumar et al. 2015), auditors (Cunningham *et al.* 2018; Du, 2017), students and instructors (Fairlie, Hoffmann, and Oreopoulos, 2014), venture capitalist syndicates (Gompers *et al.* 2016). But there is not much literature on how similarity of social identities between a manager and the board can change the executive compensation, affect agency frictions, and impacts long term firm value. This paper is an attempt to fill that gap by examining these issues.

Paper begins with gaining the intuition on how similarity of social identities (also called group identity or homophily) between a manager and the board members can change the endogenous decisions of executive compensation and firm value. To do that, I introduce the Akerlof and Kranton (2005) suggested agent utility function with an identity in a simple static model provided in Holmstrom-Milgrom (). The results provide an useful insight that the manager gets higher compensation due to in-group bias , however, this also increases the firm value because the manager puts extra effort for his group members. This implies that group identity creates an informal relationship between an agent and the principle, and reduces agency frictions in the firm.

As theoretical predictions can differ from empirical findings, therefore, it is important to check whether these theoretical predictions survive empirical analysis. To do so, first and foremost, it is essential to detect that there exist implicit biases due to the identities. According to Huettel and Kranton (2012), identities in humans can be categorized into natural and induced. Natural identities are like gender, race, caste, place of birth, native language, religion, and similar others. On the other hand, induced identities are the identities developed due to social capital like college alumni, club membership, experience based etc. Social psy-

chology and neuroscience studies has showed that the favoritism or discrimination based on natural identities are strong than induced identities and can be detected either on frequency magnetic resonance images (fMRIs) or through implicit association tests (IATs).¹ But most of these results are based on the lab experiments and therefore, hard to scale up like for this study. Therefore, I use those identities for which stereotypes are already well established in the society and these identities, in India, can learn from one single source - ‘surname’. A surname in India can provide information such as family lineage, native language, place of origin, and caste², about a person. As a result, surnames are likely an important basis of social ties and is a source of favoritism or discrimination. The surname is used as a link of social identities because of various reasons and these reasons are discussed in detail in section 3.1. But, the main reason is that within the tight kinship, effective cooperation is believed to take place within cohesive in-groups, while people outside the group are distrusted (Alesina and Giuliano 2013; Moscona, Nunn, and Robinson 2017).

Therefore, to check how group identity based on social traits can affect executive compensation, agency frictions, and long-term firm value, I develop a novel dataset on surnames mapping to their native language, caste, and native place in India. I use these social identities because implicit stereotypes or biases are stronger for these social identities in India. To form this dataset, I use the demographic data on 474 million Indians from the Socio Eco-

¹The IAT relies on test-takers’ speed of response to represent the strength of their unconscious mental associations. IATs are used to measure a wide range of implicit attitudes about social groups, products, or self identity (Greenwald et al. 1998).

²According to ancient literature, the caste is mainly divided into four - the Brahmins (priestly people), the Kshatriyas (also called Rajanyas, who were rulers, administrators and warriors), the Vaishyas (artisans, merchants, tradesmen and farmers), and Shudras (labouring classes). People who does not come in any of these castes are called ‘Harijans’ (untouchables). Brahmins, Kshatriyas, and Vaishyas are collectively called as upper caste and, Shudras and Harijans are called lower caste. After the establishment of Indian constitution in 1950, caste is redefined based on reservation provided to under-privileged in government jobs, education, and other services. As per this new categorization, Brahmins, Kshatriyas, and Vaishyas are called ‘other’ caste, Shudras as ‘schedules tribes’ (ST), and Harijans as ‘schedule caste’ (SC). The caste in SECC data is according to new categorization. I also include caste as a identity because, in India, caste remains an enduring predictor of economic status. It is correlated with occupation and employment (Prakash, 2015; Ito, 2009; Thorat and Attewell, 2007), income and expenditure (Deshpande, 2000), and capital more generally (Kijima, 2006).

economic and Caste Census (SECC) 2011 and merge it with the Linguistic Survey of India (LSI) data using the unsupervised machine learning technique. As there is no database available to provide the identities of managers and directors, therefore, surname dataset developed using big data (474 million Indians SECC data along with LSI data) is used to know the identities of managers and directors. The details of this dataset is provided in section 3.2. These identities related to surname forms the identities of the principal (board) and an agent (manager).³

By discerning the identities of managers and directors, I develop the homophily variables related to native language, native place, and the caste, and examine their association with executive compensation, firm value, and residual firm value (firm value conditioned on executive compensation). Residual firm value explains the reciprocation of in-group favoritism by the manager.

The main empirical results show that the manager gets higher total compensation due to the in-group favoritism based on native language and native place, and the long term firm value (measured as Tobin's Q) and residual firm value (proxy for manager's reciprocation to in-group favor) are higher for such firms compared to others. These results are robust to firm fundamentals and corporate governance characteristics.

To check the external validity of homophily constructs, I develop the identity variables based on alternative dataset from "Indianchildnames.com". This website features a largest collection of about 125,000 names, 15,000 surnames of Indian and International origins. Using this new dataset, I again develop homophily variables based on native place and native language. The results are robust to this test. To conduct the external validity test of dependent variables, the total compensation is replaced with the salary and the Tobin's Q with the firm's sales growth. The results remain consistent that the manager gets higher

³The surname dataset provides the composition of all the Indian surnames at state level, language level, and at caste level. As there is no major intra-country movement of any community in India except Kashmiri pandits in 1990, this surname database would provide correct identities of managers and of directors.

compensation due to in-group favoritism and the group identity reduces agency frictions and increases the long-term firm value.

As there can be endogeneity because the board can select a manager who has the same identity as their own to reduce the frictions in corporate decision making (Damaraaju and Makhija, 2018) and thus, this can reduce the impact of the main results of the paper. To mitigate this concern, I use similar identification strategy as in Fisman, Paravisini, and Vig (2017). I use the rotation of independent director(s) within the same gender as a shock to the identity of the board. The idea here is to keep the corporate governance characteristics same and give a shock to the identity of the board. Using this shock to board identity, I find that results are robust to endogeneity concerns.

Lastly, I test for parallel conjectures such as influence of powerful manager in the firm and college ties or ties from past employment between a manager and board members . Adams et al. (2005) show that firm in which CEO has greater power to influence decisions also has higher performance variability. Therefore, I control for 'powerful CEO' measures of Adams et al. (2005) and show that homophily results still holds. Farcassi and Tate (2012) show that powerful CEO are more likely to appoint known directors and this CEO-director ties reduce firm value. From these network connections, the most common are past employment and the least common are education and current employment. To mitigate the concerns that results are driven by ties from past employment or college ties, I control the measures based on these ties. The homophily results are still robust. The coefficient of 'powerful CEO' measures and past experience ties are in line with these papers.

In sum, this paper shows that shared group identity between the board members and the manager can increase the executive compensation due to in-group favoritism and the firms with group identity have higher value compare to other firms. Furthermore, the residual firm value (i.e. the firm value conditional on executive compensation of the last year) is higher for such firms, implies that manager puts extra effort when there is shared group identity

between the board and the manager. These results imply that although in-group favoritism is a cost, however, it can reduce agency frictions and can increase long-term firm value.

Related Literature: This paper contributes toward the literature on executive compensation, corporate governance, and the role of identity economics in corporate finance.

The literature on executive compensation is rich and there are many papers on executive compensation based on cross-sectional variation (Roberts (1956); Baker, Jensen, and Murphy (1988); Barro and Barro (1990); Murphy (1999); Gabaix and Landier (2008); Frydman and Saks (2010); and Gabaix, Landier, and Sauvagnat (2014)), value of pay (Lambert, Larcker and Verrecchia (1991); Dahiya and Yermack (2008)), structure of pay (Frydman and Saks (2010)), level of pay (Gabaix and Landier (2008); Frydman and Saks (2010)), sensitivity of pay (Conyon, Core, and Guay (2011)), rent extraction view ((Bertrand and Mullainathan (2001); Bebchuk, Fried, and Walker (2002); Bebchuk and Fried (2003)), and the effects of executive compensation (Morck, Shleifer, and Vishny (1988); Mehran, (1995); Kim and Lu (2011)). This paper fits in the last two themes of executive compensation. The in-group favoritism by the board toward a manager of their own identity can be considered as rent extraction (or hidden compensation) and it is also important to know how this affects the firm value in the long run.

Corporate governance and its effectiveness plays a big role in executive compensation (Core, Holthausen, and Larcker, 1999; Fahlenbrach, 2009; Armstrong, Ittner, and Larcker, 2012). Researchers have checked the impact of board size, number of independent directors, gray directors and staggered board on executive compensation. But the identities of the board members are also important. Delis *et al.* (2017) show that the impact of genetic diversity in the country of origin of the firms' board members on corporate performance and find that adding board of directors from countries with different levels of genetic diversity

(either higher or lower) can increase firm performance. Using a sample of US firms and gender as identity, Adams and Ferreira (2009) find a negative impact of having females on the board on firm performance, despite better attendance records and more effective monitoring in firms with more gender-balanced boards. In contrary, Gregory-Smith et al. (2014) find no evidence that the gender composition of the board affects firm performance. Core, Holthausen, and Larcker (1999) showed that firms with weaker governance structures have greater agency problems and CEOs at such firms receive greater compensation; and that firms with greater agency problems perform worse. In this paper, I examine how a shared identity between the board and the manager can affect agency frictions, and hence, the CEO compensation and future firm performance.

Lastly, this paper contributes to the literature on identity economics in corporate finance. The identity economics literature mainly discusses how identities can change outcomes due to the implicit biases or due to the taste-based discrimination (Becker, 1971). Identities can be divide into two types - by experience (educational institute, past employer, same club etc) or by birth (gender, native place, native language, race, caste etc.). The implicit biases and stereotypes are much stronger in the latter case. Like, Fisman *et al.* (2018) shows how hometown ties of a candidate with fellow selection committee members can increase the selection probability in Chinese Academies of Sciences and Engineering by 39%. Fisman, Paravisni, and Vig (2017) show how group identity between the loan officer and the borrower based on the identities like religion and caste increases credit access and loan size dispersion, reduces collateral requirements, and induces better repayment even after the in-group officer leaves. Contributing to this literature, this paper examines how the similarity in identities (based on caste, native place, and native language) between board and manager can change managerial compensation, managerial efforts towards the firm, frictions between a manager and the board and the future firm performance.

Structure of the Paper: The rest of this paper proceeds as follows. Section 2 provides an intuition on how the similarity of agent’s identity with the board can change the compensation and the firm value. Section 3 explains the datasets and the variables used for the analysis. Section 4 explains methodology used for surnames matching and the empirical specification used for the analysis. Section 5 provides the perspective on endogeneity concerns and the identification strategy used to mitigate those concerns. Section 6 discusses the results. Finally, Section 7 provides the concluding remarks.

2 Principal - Agent Problem with Identity Function

For this work, I use principal-agent model provided in Holmstrom (197) and Edmans, Gabaix, and Jenter (2017) for the executive compensation setting. The board of directors hires a manager to run the firm. The firm value is given by $V(a, S, \epsilon)$ which is increasing in the manager's action a and firm size S . Suppressing the dependence on S and ϵ for simplicity.

$$V(a) = S + b(S)a + \epsilon$$

The manager is paid a compensation $c(V)$ contingent upon firm value. The limited liability on the board is assumed which means $c(V) \leq V$; that is manager pay should not be more than firm value. Similarly, the limited liability on manager is assumed which means $c(V) \geq 0$ and reservation wage, $w \geq 0$.

The modified principal problem with group identity is shown below. The group identity is introduced in the preference function by providing different weights to the managerial effort component. This defines the agent utility function with identity and it implies that the agent exerts more effort if the principal (board, in this case) shares the same identity as the agent. This kind of agent utility function is suggested by Akerlof and Kranton (2005). The similar way of introducing the group identity in preferences is also adopted in Chen and Li (2009) and Chen and Chen (2011) and is related to the literature of minimum-effort coordination games.⁴

⁴For more details on this topic, refer to Camerer (2003).

The simple principal problem in static world from Holmstrom and Milgrom (1979) can be seen as,⁵

$$\begin{aligned}
& \max E[V-c] \\
& \text{s.t. } E[c - g(a)] \geq u(w) \quad (\text{Participatory Constraint}) \\
& c = \phi + \theta V \\
& a^* \in \arg\max_a E[c - g(a)] \\
& V = s + b(s)a + \epsilon
\end{aligned}$$

$g(a)$: cost function of manager's efforts

ϕ : Fixed component of the manager's compensation (Salary).

θ : Variable component of the manager's compensation

ϵ : Random variable with normal distribution. $\epsilon \sim N(0, \sigma^2)$

Agent utility function from Akerlof and Kranton (2005),

$$U(c, a, m) = c - a + I_m - t_m |\hat{a} - a|$$

$\hat{a}$: Ideal or optimal effort

I_m : Additional utility due to group identity (or homophily)

Agent's effort including identity $= a - t_m(\hat{a} - a)$

m : Indicator variable for homophily (or group identity)

Firm value and manager's compensation using his efforts including identity,

$$V = S + b(s)[a - t_m(\hat{a} - a)] + \epsilon$$

⁵For simplicity, the terms without the exponential part are showed. For details, refer Edmans, Gabaix, and Jenter (2017, Pp-43-45).

$$c = \phi + \theta[s + b(s)[a - t_m(\hat{a} - a)]]$$

Following Holmstrom and Milgrom (1979), I use quadratic cost function of manager's efforts i.e

$$g(a) = \frac{1}{2}ga^2$$

$$E(u) = E[c - \frac{1}{2}g[a - t_m(\hat{a} - a)]^2]$$

First order condition (FOC) :

$$\frac{dE[u]}{da} = \theta b(s)(1 - t_m) - g[a - t_m(\hat{a} - a)](1 - t_m)$$

$$\frac{\theta b(s)}{g} = a^* - t_m(\hat{a} - a^*)$$

if $m = 1$, the board and a manager share the same identity then $\hat{a} = a^*$ and therefore,

$$a^* = \frac{\theta b(s)}{g} \tag{1}$$

Else if $0 \leq m < 1$, that is board and manager have different identity, then

$$\frac{\theta b(s)}{g} = a - t_m(\hat{a} - a)$$

$$a^* = \frac{1}{1-t_m} \left[\frac{\theta b(s)}{g} - t_m \hat{a} \right] \tag{2}$$

From participation constraint,

$$E[c] = \begin{cases} w + \frac{1}{2}g\left(\frac{\theta b(s)}{g}\right)^2 - \frac{\nu}{2}\theta^2\sigma^2, & \text{if } m = 1 \\ w + \frac{1}{2}g\frac{1}{(1-t_m)^2} \left(\left[\frac{\theta b(s)}{g} - t_m \hat{a} \right]^2 - \frac{\nu}{2}\theta^2\sigma^2 \right), & \text{if } 0 \leq m < 1 \end{cases} \tag{3}$$

Principal's utility = $E[V - c]$

$$= S + b(s)a^* - E[c]$$

First Order Condition (FOC) with respect to θ ,

$$\theta^* = \begin{cases} \frac{1}{1+g\nu(\frac{\sigma}{b(s)})^2}, & \text{if } m = 1 \\ \frac{1-t_m[1-\frac{g\hat{a}}{(b(s))^2}]}{1+g\nu(\frac{\sigma(1-t_m)}{b(s)})^2}, & \text{if } 0 \leq m < 1 \end{cases} \quad (4)$$

From participation constraint,

$$\phi^* = w - \theta^*S - \theta^*b(s)(a^* + t_m(\hat{a} - a^*)) - \frac{1}{2}g(a^*)^2 - \frac{\nu}{2}(\theta\sigma)^2 \quad (5)$$

a^* and θ^* are known from eq(1), eq(2), and eq(4)

As from eq(1) and eq(2), $a_{(m=1)}^* > a_{(0 \leq m < 1)}^*$

Therefore, at equilibrium,

$$V_{(m=1)} > V_{(0 \leq m < 1)}$$

This implies that the firm in which manager and board share the same identity has higher value than the similar other firm.

From eq(4),

$$\theta_{(m=1)}^* > \theta_{(0 \leq m < 1)}^*$$

It shows that fixed component of manager’s compensation is higher for firm with group identity. As variable component is function of firm value and the firm value is higher for a firm with group identity, hence, total compensation of the manager is higher for a firm with group identity.

In sum, total compensation (and salary) of the manager and firm value are higher if the manager and the board share the same identity (i.e. group identity or homophily). Although, additional managerial compensation is a cost to the firm due to in-group favoritism, this cost is compensated by manager’s extra efforts and thus, by increasing the firm value. In the next section, I verify whether these theoretical predictions using the empirical analysis.

3 Data

3.1 Surname Data

To know the identities of managers and the directors, I use surname as the source. The surname is used as a link of social identities because of various reasons. Sociological studies find that “surname remains central to the establishment of a person’s sense of belonging to a kin group” in contemporary times (Davies, 2011) and anthropologists have long noted that kinship systems differ in their tightness: the extent to which people are embedded in very large extended family networks. With tight kinship, effective cooperation is believed to take place within cohesive in-groups, while people outside the group are distrusted (Alesina and Giuliano 2013; Moscona, Nunn, and Robinson 2017). Conversely, in loose kinship societies, people are said to enter productive interactions with strangers but do not place special emphasis on the in-group member. Surname is widely perceived as an important means of genealogy, potentially revealing one’s DNA information (Jobling, 2001). It may also indicate one’s ethnic minority background and engender race biases (Kumar et al., 2015).

Importantly, a surname connects the bearer not only to family members, but also potentially to many non-family members who share the same surname (Carr and Landa, 1983; Kiong and Kee, 1998). This creates a network and a bond embedded with rich social, cultural or even biological connections. These connections can be invoked when people bearing the same surname meet or work together, making it easier for them to understand and cooperate with each other. A surname in India can provide information such as family lineage, native language, native place, and caste.

As three identities - native language, native place, and caste can be learn from a surname using Socio Economic and Caste Census (SECC) data and Linguistic Survey of India (LSI) data, I develop a surname database mapped with these identities. The surname database used to know the identities of managers and the directors.

3.2 Microcensus Data

To generate identity variables based on native place and the caste, I use the Socio Economic and Caste Census (SECC) 2011 dataset of 474 million Indians.⁶ This Census is unique because in the history of independent India, only two times caste is measured - in 1950 and in 2011. The Registrar General of India conducted SECC 2011 and has 46,73,034 categories of caste, sub-caste, synonyms, different surnames, gotras in the caste and clan names.⁷ The data contains individual name, age, gender, address (state, district, and gram panchayat (similar to county)), caste, parents' name and other details. To keep individuals' information private, I extract only surnames, caste, district, and state.

The extracting surnames from Indian names using SECC data is a complex process due to following reasons: First, names in the SECC data are given in English as well as in native language seperated by backslash ("") Second, many time names are without surnames or first

⁶The exact number of observations in raw SECC data are 473,967,954.

⁷<https://www.thehindu.com/opinion/lead/the-tools-for-counting/article24247791.ece>

name and surname are amalgamated such as "DilipChauhan" (surname: Chauhan). When this occurs with surnames of same character length as only first name or amalgamated first-last name, it makes difficult to extract surnames without including the noise. Third, names in north India and south India are differently written in SECC data such as "Ram Singh" is a north Indian name with surname 'Singh' and 'Chandra Rao P.K.' is a south Indian name with surname 'Rao'. It exhibits that there is a different position of surnames in north Indian and south Indian names in SECC data and this makes extraction further complicated. The long names such as 'Mohammad Kasim P.' (surname: Kasim) also have the same pattern as south Indian names. The details of generating surname database from SECC 2011 using cloud computing is explained in **Appendix C**. The final dataset is consolidated at the state level and contains variables like surname, caste, and place of origin. I consolidated the data at the state level because stereotypes and implicit biases are stronger at the state level rather than at the district level.⁸

3.2.1 Linguistic Survey of India (LSI) Dataset

The LSI is a comprehensive survey of Indian languages conducted for the first time in 1898 by an linguist George Abraham Grierson, a linguist, and obtained information on 364 languages and dialects. The second survey was conducted in 1984 and by 2010; it was 40% complete but the project was abandoned after that. The third survey started as People's Linguistic Survey of India (PLSI) in 2010. According to 1991 census of India, there are about 1,576 "mother tongues" with separate grammatical structures and 1,796 languages classified as "other mother tongues."

For this paper, I use Lacina (2017) data which contains linguistic composition of all the 640 districts of India. This data contains information on district id, state, district name, population, Hindu population in a district, and the seven most popular languages in a dis-

⁸Python code to scrape surnames with identities such as caste and state is available on request.

trict with a percentage of population speaking them.

For simplicity, I use only the most popular language in a district and consolidated the whole data at the state level. This provides data on the most popular language in 29 states and 7 union territories. Final surname dataset contains native place, native language, and the caste associated with those Indian surnames.

3.2.2 Alternative Surname Data

I develop the alternate dataset from the website-“Indianchildnames.com”. This website features a large collection of about 125,000 names and 15,000 surnames of Indian and international origins. Data on Indian surnames, their native language, and their native place is collected using a web scraping technique in R. This dataset provides information on unique 2,225 Indian surnames with their native language and native place. Merging this data with board data of ProwessDx (CMIE) based on surnames gives information on language and place of origin of directors and managers. As there is no common identifier between the surnames dataset and the board data of ProwessDx , therefore, a text similarity algorithm is used to accomplish this task. The details of this algorithm is provided in section 4.1.

3.3 Firm characteristics and Manager characteristics

ProwessDx (CMIE) and the BoardEx databases used for manager and board members name, executive compensation data, firm fundamentals, and board characteristics. The extracting surnames from manager and board members name is not straightforward because of various designations and their patterns such as Mr./MR/Mr/(MR)/(Mr.)/(Mr), Dr./Dr/(Dr), Prof., Sh./Shri, I.A.S., army designations (Maj. (Major), Gen. (General), Col. (Colonel), navy designations (Vice Admiral, Commodore) and many others. Therefore, an algorithm used to extract surnames from a manager and the board members names is

slightly different from the way surnames extracted from SECC names.

To avoid data complexities or unavailability of firm fundamentals, sample comprises only non-financial firms listed on two main stock exchanges - Bombay Stock Exchange (BSE) and National Stock Exchange (NSE), from 2004-2018. There are 3,668 firms in the sample and 41,507 firm-year observations. **Appendix A1** shows the filtration of observations from ProwessDx database to sample data. As concept of CEO is new in India, therefore, CEO-equivalent positions like Managing Director, President and Chairman are also kept in the sample.⁹The distribution of managerial designation is provided in **Appendix B**. The yearly distribution of the observations shows that the sample is balanced(**Appendix A2**).

The compensation data from ProwessDx provides the information on total compensation, salary, perks, and retirement benefits. For the analysis, only total compensation and salary data is used as most of the observations for perks and retirement benefits data is missing. As total compensation and salary is skewed, therefore, log values of these variables are used.

Following Morck, Shleifer, and Vishny (1988) Tobin's Q (*TobinQ*) is used as proxy for firm value. The *TobinQ* is measured as a ratio of market price of the equity value by book value of the equity value. For the robustness check, *Sales Growth* is used as another proxy of firm value. *Sales Growth* is calculated as growth in sales compared to the previous year.

For the controls, *Size*, *Firm age*, *Institutional Ownership*, *Promoters Ownership*, *Leverage*, and *ROA* are used. The data to construct these variables are from ProwessDx (CMIE). *Size* is measured as log of market capitalization. *Firm Age* is the age of the firm from year of incorporation. *Promoters Ownership* is percentage of promoters equity holding to the total equity issued by the firm. *Institutional Ownership* is percentage of institutional investors equity holding to the total equity issued by the firm. *Leverage* is the ratio of total debt by total assets. *ROA* is measured as ratio of net income to total assets.

Because manager characteristics like age, gender and education are not available in

⁹Bertrand and Hallock (2001) uses the CEO, chairman, president positions for the top corporate jobs.

ProwessDx, therefore, this data is gathered from BoardEx database. But due to constraints in matching between these two databases, these characteristics are not available for all the managers. Due to the same issue, the observations of board characteristics such as board diversity, board size, and board independence are limited to 4,451. The results controlling for board characteristics are provided in **Appendix B**. For the main regression, the institutional ownership and promoters ownership are used as proxies of governance.

4 Methodology

4.1 Surname Matching using Unsupervised Machine Learning

Due to lack of common identifier between surname dataset and ProwessDx dataset, to know the identities of a manager and the remaining board members, a text similarity algorithm is used. Text similarity algorithm is a simple form of unsupervised machine learning. The two most common text similarity algorithms are Jaccard and Cosine. These two algorithms gives different results if the two documents or sentences in question are too different in length. In this case, both should give similar results as comparison is between two surnames rather than between not documents.

For matching, surnames are first divided into bi-grams and by using Jaccard similarity algorithm to calculate a similarity scores are computed. For this case, the Jaccard similarity is the ratio of the number of common bi-grams in two surnames divided by the union of bi-grams in two surnames.

$$\text{Jaccard Similarity} = \frac{A \cap B}{A \cup B} \quad (1)$$

Here A and B are two surnames.

It can be understood by computing the Jaccard similarity score of ‘smith’ and ‘thomas’. The bi-grams of ‘smith’ are {sm,mi,it,th} and bi-grams of ‘thomas’ are {th,ho,om,ma,as}. As there is only one bi-gram matching i.e. ‘th’, therefore, Jaccard similarity is 0.125. It is quite low because the surnames in this example are too different. For exact matching, the threshold of similarity score is kept at 0.9 and above, and to confirm the matching, the manual check is also conducted.

4.2 Empirical Specification

To test the role of in-group favoritism in setting the executive pay, following model is used:

$$Y_{i,t} = \alpha_{i,t} + \beta_1 Homophily_{i,t} + \beta_2 X_{i,t} + \beta_3 L_{i,t} + \beta_4 Z_{i,t} + \gamma_{firm} + \gamma_{year} + \gamma_{Designation} + \epsilon_{i,t} \quad (15)$$

The main dependent variable, Y, is either salary or the total compensation. Homophily (measurement of group identity) is a dummy variable which takes value 1 if the majority of board members have the same identity traits (same native language, same caste, or same place of origin) as executive/manager; otherwise 0. This paper uses a simple baseline homophily measure as an estimate of in-group favoritism.¹⁰ The detailed literature surveys

¹⁰The extensive literature on homophily has demonstrated such patterns across many dimensions, including ethnicity, age, gender, religion, profession, as well as things like political opinions and other behaviors.¹ Homophily can arise for many reasons, such as biases in the amount of contact that people have with each other (Blau (1977), Feld (1981), Rytina and Morgan (1982)), preferences for associating with individuals with similar traits, behaviors, or backgrounds (Cohen (1977), Kandel (1978), Knoke (1990), and Currarini, Jackson and Pin (2008) and Bramoulle and Rogers (2009)), competition among groups (Giles and Evans (1986)), social norms and culture (Carley 1991)), and institutional and organizational pressures (Meeker and Weiler (1970), Khmelkov and Hallinan (1999), Kubitschek and Hallinan (1998), Stearns (2004)).

on executive compensation by Core, Guay, and Larcker (2002), Frydman and Jenter (2010), and Edmans, Gabaix, and Jenter (2017) suggest that the main determinants of executive pay are mainly firm fundamentals, governance characteristics, and CEO characteristics. Following that suggestion, X is a vector of firm characteristics, L is the vector of governance characteristics (such as board size, number of independent board members, and number of female board members), and Z is the vector of executive characteristics (such as education, age, gender, and others). The γ_{firm} and γ_{year} are firm and year fixed effects. As results can be affected due to managerial positions, therefore, $\gamma_{Designation}$ fixed effects is also used.¹¹

To check the impact of homophily on the firm value, following empirical specification is implemented:

$$TobinQ_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

The coefficient of interest is β_1 . As homophily is based on language, place, and caste, β_1 can be seen as $\beta_{Language}$, β_{Place} , and β_{Caste} . *Size*, *Firm Age*, and *Leverage* are controls for firm fundamentals. *Institutional Ownership* and *Promoters Ownership* are proxies of governance. Based on the prior literature, all β s of firm fundamentals and governance are expected to be positive except β_6 . To avoid any interference of firm level unobservables and time trends, firm and year fixed effects are also used.

¹¹Through out the paper, I mainly control for firm fundamentals and use institutional investors holding and promoters holding as a proxy for governance. However, I also show the results using governance characteristics like board size, board diversity, and board independence in the **Appendix D**. As the manager characteristics such as age, gender, and education are available only in BoardEx data and not in ProwessDx data, and matching these two data sets at individual level on the name based ended with very few observations, therefore, I did not show the regression results controlling for the manager characteristics.

Next, to check the reciprocation of in-group favor by the manager, the impact of homophily on the residual firm value is examined. The residual firm value is firm value conditioned on managerial compensation of last year and it is measured as Residual Q.

$$TobinQ_{i,t} = \xi_i + \phi TotalCompensation_{i,t-1} + \varepsilon_{i,t}$$

The Residual Q is ε . It captures the firm value conditioned on managerial compensation; an increase in Residual Q proxies higher managerial efforts. . To test whether Residual Q is positively associated with homophily variables or not, following regression specification is used,

$$ResidualQ_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

The interpretation and the expectations for the β s is same as for the firm value regression.

5 Endogeneity Concerns and Identification Strategy

As board can select the manager of the same identity in order to reduce frictions in corporate decision making (Damaraju and Makhija, 2018), and this can affect the causal inference of the results of this paper. Therefore, to mitigate this concern, I use similar identification strategy as in Fisman, Paravisini, and Vig (2017). The rotation of independent director within the same gender can be considered as an exogenous shock because it is only changing the identity of the overall board by maintaining the board characteristics such as board size, board independence, and the board diversity.

As ProwessDx does not have information on the board members' characteristics like age,

gender, and similar others. Therefore, first, the dataset on this shock is created using the Boardex data ¹² and then merged it with ProwessDx sample using ISIN and year.

As there is a limited mapping between ProwessDx and Boardex datasets, therefore, I could able to match 4,205 observations. The independent director rotation dummy (*ID Rotation*) constructed which takes value 1 if the there is a replacement of independent director with the same gender but with different identities. The empirical specification for this test is,

$$Y_{i,t} = \alpha_i + \beta_1 IDRotationXHomophily_{i,t} + \beta X_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

Here Y is total compensation (*Total Compensation*), firm performance (*TobinQ*), or firm performance conditioned on managerial compensation (*ResidualQ*). The coefficient of interest is β_1 . The firm (γ_{firm}) and year (γ_{year}) are used to control for firm and time unobservables.

¹²Boardex dataset on India uses the various sources of information like specific database of Indian directors (Directors Database), media information, etc. to construct their dataset.

6 Results and Discussion

6.1 Univariate Analysis

6.1.1 Identities of the Managers and the Directors

The Figure 1 shows the distribution of managers and directors as per their native place. These maps showcase that although the directors and the managers are from all over India but most of them originally from southern India. In a similar analysis, I find that most of the founders comes from northern India. This implies that, in India, wealth creators come from north region while wealth managers come from south region.¹³

[Insert Figure 1 Here]

Figure 1 also shows that most of the directors and the managers in India belong to the upper caste and only 21-24 % are from a lower caste. This distribution is more skewed at managerial level than at directors level.

Figure 2 shows the distribution of managers and directors as per their native language. It shows that most of the managers and directors speak Marathi, Tamil, or Western Hindi and this group comprises 14-18 % of the sample. Other popular native languages spoken by managers and directors are Malyalam, Gujarati, and Kannada. The native place, native language, and caste forms the identities of the managers and the directors.

[Insert Figure 2 Here]

¹³Indian Media has also pointed out this that the half of the top ten Indian billionaires figuring in the list of Bloomberg Billionaires are Gujaratis (state in a northern India) and one in three CFOs of Nifty index companies is south Indian. For details check, <https://economictimes.indiatimes.com/news/politics-and-nation/stereotypes-dominate-indias-rich-list/articleshow/46857906.cms?from=mdr>

6.1.2 Summary Statistics

Table 1 shows the summary statistics of the main variables used in the analysis. The average (maximum) total compensation of CEO or CEO equivalent position in India is 144,180 USD (2,024,083 USD).¹⁴ Major part of the total compensation is salary and the variable components of the pay comprises perquisites and retirement benefits. The average *Salary* is 101,644 USD and maximum is 965,457 USD. Unlike US, the mean variable pay in India is 17% of total remuneration. To make sure these numbers are correct, I compare distribution of managerial compensation and summary statistics with Chakrabarti et al. (2012). As total compensation and salary are skewed, therefore, I use the log of total remuneration and salary. Figure 3 shows the kernel density curves of *Log Remuneration* and *Log Salary* with normal density curves for reference. It shows that using the logarithmic values of total remuneration and salary reduced the skewness in these variables.

[Insert Figure 3 Here]

The mean (max) of Log Remuneration (*TotalCompensation*) is 10.239 (13.634). *Log Salary* is log of the salary component. The mean(max) of *Log Salary* is 10.7 (13.493). The standard deviation of total compensation of the manager is higher than the standard deviation of the salary component and this comes from the perks. These two variables are used to show how the executive compensation gets affected by the in-group favoritism.

[Insert Table 1 Here]

To show how the firm value is affected by the group identity (homophily), *Tobin's Q* and *Sales Growth* are used as proxies of firm value. *Tobin's Q* is the ratio of market price of a equity to the book value of the equity. The mean (max) value of *Tobin's Q* is 2.231 (19.260).

¹⁴To provide the sense of these numbers in comparison to US executive compensation, for the period 1982 through 1988, the average salary and bonus for CEOs of large public companies was 843,000 USD. (Jensen and Murphy, 1990)

Sales Growth is change in sales compared to last year. The mean (max) of *Sales Growth* is 0.145(7.067). To show how the firm value is affected by the homophily even after paying the cost of in-group favoritism, *Residual Q* is computed. The *Residual Q* is the residual of regressing the firm value on last year's managerial compensation. The mean (max) of *Residual Q* is -0.019 (8.470).

The three main independent variables are *Homophily Language*, *Homophily Place*, and *Homophily Caste*. These variables are dummy variables and take value 1 if the board and the manager share the same identity i.e. native language, native place, or caste. In the 44-46% of the observations in the sample, the board and the manager speaks the same native language or belongs to same native place. This indicates that the sample is balanced on these identities. However, there is a skewness in the variable - *Homophily Caste*. In 90 % of the observations, the board and the manager belong to the same caste. As the board and the managerial positions are dominated by the upper caste individuals, therefore, the skewness in *Homophily Caste* results from it. Table 1 Panel B shows the distribution of these Homophily variables in further detail.

The control variables used are *Size*, *Firm Age*, *Leverage*, *Business Group*, *Promoters Ownership*, *Promoter-Manager*, *Institutional Ownership*, *Board Independence*, *Gender Diversity*, and *Board Size*. The last three variables are estimated from BoardEx database and the remaining control variables are from ProwessDx database. The *Size* is measured as log of market capitalization. The mean *Size* is 16.982. The average firm age in the sample is 32 years and the average leverage of the firm is 32.5 %. As in India, the promoters own majority of shares and it is evident here. The *Promoters Ownership* on average is 53 %. In most of the firms in India, one of the promoters keep a managerial position. In this sample, such cases are 93 %. The *Institutional Ownership*, on average, is 8 % . The definition of all the variables are also provided at the end of the paper.

Regarding board characteristics, the average board size is 11 members. The women in

board (*Gender Diversity*) is only 9-10 % on average. The mean board independence is 50 % in the sample.

6.2 Main Results

6.2.1 Executive Compensation and In-Group Favoritism

The Table 2 shows how managerial compensation changes in-group favoritism when the manager and the board share the same identity. The regression specification to test this is,

$$TotalCompensation_{i,t} = \alpha_i + \beta Homophily_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

The results show that when the manager and the board share the same native language or come from same native place or both, it helps the manager to earn higher compensation. The impact of group identity is stronger for native language than native place as $\beta_{Language} > \beta_{Place}$. The $\beta_{Language}$ is 0.052 and the β_{Place} is 0.035. But when the board and the manager have the same caste, this type of homophily affects the total compensation negatively. This result is against the expectation but this could be due to skewness in the caste variable and therefore, this requires further analysis. To control the impact of unobservables, managerial designation, firm, and year fixed effects are used.

[Insert Table 2 and Table 3 Here]

Table 3 shows the impact of group identity on the executive compensation with controls. The regression specification for this is,

$$TotalCompensation_{i,t} = \alpha + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 ROA_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

These results confirm the previous results that if the manager and the board speaks same native language or belong to same native place then the manager earns higher compensation compared to his counterparts. The $\beta_{Language}$ is 0.140 and it is significant at 5% level. The β_{Place} is 0.077 and it is significant at 10 % level. This implies that $\beta_{Language} > \beta_{Place}$. To provide the insight on economic significance of these results, 28.6 % increase in homophily based on language can increase the total compensation by 10,000 USD (6.93 % of mean total compensation). As suspected, homophily caste variable is insignificant in the presence of controls. Among the controls, Size, ROA, Institutional Ownership, and Promoters Ownership are significant and positively associated with managerial compensation. The latter results are align with the results in executive compensation literature.

6.2.2 Firm Value and Group Identity

Table 4 shows the impact of homophily on firm value. The regression specification used for this purpose is,

$$TobinQ_{i,t} = \alpha + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

The results show that homophily in native language or in native place are positively associated with TobinQ and the results are significant at 5% level. Similar to executive compensation results, homophily based on native language is stronger than homophily based on native place, i.e. $\beta_{Language} > \beta_{Place}$. The $\beta_{Language}$ is 0.205 and it is significant at 5% level. The β_{Place} is 0.168 and it is also significant at 5 % level. Among the controls, The Size, Institutional Ownership, and Promoters Ownership are significant at 5 % level and are positively associated with TobinQ. Leverage is also significant at 5 % level and the negatively associated with TobinQ. To avoid any time trend and firm level unobservables, I also use

firm and year fixed effects.

[Insert Table 4 Here]

6.2.3 Firm Value and Reciprocation of In-Group Favoritism

Table 5 shows the results of group identity on Residual Q. The Residual Q is residual from the regression of Tobin Q on managerial compensation of last year.

$$TobinQ_{i,t} = \xi + \phi TotalCompensation_{i,t-1} + \varepsilon_{i,t}$$

The Residual Q is ε . It captures the firm value conditioned on managerial compensation and increases in this value should come from higher managerial efforts. Therefore, I check whether Residual Q is positively associated with homophily variables or not. To test this, I use following regression specification,

$$ResidualQ_{i,t} = \alpha + \beta_1 Homophily_{i,t} + \beta_2 Size + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

The results show that homophily is positively associated with Residual Q. Similar to previous results, the impact of homophily based on native language is stronger than homophily based on native place, i.e. $\beta_{Language} > \beta_{Place}$. The $\beta_{Language}$ is 0.148 and it is significant at 5 % level. The β_{Place} is 0.139 and it is also significant at 5 % level. The homophily due to caste is insignificant.

[Insert Table 5 Here]

6.3 External Validity Test

6.3.1 Alternative Compensation and Firm Performance Variable

To make sure that these results are not due to a particular dependent variable for executive compensation and firm performance, log total compensation is replaced with log of salary and firm performance is replaced with Sales Growth. The empirical specifications for these tests are,

$$Salary_{i,t} = \alpha + \beta_1 Homophily_{i,t} + \beta_2 Size + \beta_3 FirmAge_{i,t} + \beta_4 Institutional_Ownership_{i,t} + \beta_5 Promoters_Ownership_{i,t} + \beta_6 ROA + \gamma_{Designation} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

$$Sales_Growth_{i,t} = \alpha + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 Institutional_Ownership_{i,t} + \beta_5 Promoters_Ownership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_t$$

The results of these tests (shown in Table 6 Panel A) are in line with the previous results ; in-group favoritism based on native language and on native place helps a manager to earn higher salary and the firms with group identity have higher sales growth compare to other firms. This strengthens the main argument of the paper that although group identity leads to in-group favoritism, this also reduces the frictions and therefore, bearing this cost of in-group favoritism is worth for the firm.

6.3.2 Alternative Homophily Variable

To check the external validity of the homophily variables, rather than using the SECC 2011 and LSI data, the data from Indianchildnames.com is used. The empirical specification for this test is,

$$TotalCompensation_{i,t} = \alpha + \beta_1 In-Group-Favoritism_{i,t} + \beta_2 Size_{i,t} + \beta_3 ROA + \beta_4 Leverage_{i,t} + \beta_5 Board_Size_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

Here *TotalCompensation* is either total compensation or the salary and *In-Group-Favoritism* is a group identity dummy based on native language. In Table 6 Panel B, The coefficient of *In-Group-Favoritism* is 0.138 in the regression of total compensation and 0.102 in regression of the salary. In both cases, results are significant at 5 % level.¹⁵ To avoid the effect of firm level unobservables or any time trend, I use firm fixed effects and year fixed effects.

[Insert Table 6 Here]

These results confirm that if the manager and the board share the same identity, then the manager and the firm both are better off as it reduces agency frictions.

6.4 Endogeneity Issue and the Identification Test

The endogenous relationship between executive compensation and corporate governance is always an issue (Hermalin and Weisbach, 1998 ; Wintoki et al. 2012) for this kind of research question. Recall, there is a possibility that the board chooses a manager that matches their own identity just to avoid conflict in corporate decision making (Damaraju and Makhija, 2018). Thus to mitigate such concerns, the rotation of independent director within the same gender is used as a shock to the board identity.

The empirical specification for this test is,

$$Y_{i,t} = \alpha_i + \beta_1 IDRotationXHomophily_{i,t} + \beta X_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

¹⁵I also check the results using the group identity dummy based on native place and the impact of group identity on firm performance. Results supports the main hypothesis of the paper.

Here Y is total compensation (*Total Compensation*), firm performance (*TobinQ*), or firm performance conditioned on managerial compensation (*ResidualQ*).

[Insert Table 7 Here]

I find that β_1 is positive in all the three regression specification and but statistically significant for firm value and for residual firm value specification . This confirms the main results of the paper that group identity between the manager and the board leads to higher compensation to the manager and but this also reduces agency frictions. Thus, although there is a cost of in-group favoritism paid by the firm, however, the same incentivizes the manager to put more efforts toward the growth of the firm and increase its long term value.

6.5 Other Robustness Tests

6.5.1 Community Migration and Group Identity

The underlying assumption of using the surname data to know the identities of managers and directors is that there is no migration of whole community (people belong to particular surname) from one part of India to another. For example, if majority of 'Gupta' community decided to live at Bengal after Independence, it means majority of 'Gupta' community still reside there. This premise is quintessential because change in place would also affect the linguistic profile and the caste profile related to that surname and hence, the identities of manager and directors. Although this premise is valid for most of the communities (or surnames) but not true for all. The community such as 'Kashmiri Pandits' migrated from northern most part of India known as Kashmir region to other parts of India due to the riots in 1990s in that state.¹⁶ As it is hard to track each 'Kashmiri Pandit' surname and exclude them from surname database, therefore, I exclude all the surnames which are associated with Kashmir region and do the robust check with remaining surnames.¹⁷

¹⁶<https://www.bbc.com/news/world-asia-india-35923237>

¹⁷318 observations with surnames from Kashmir region excluded from the sample to conduct this test.

[Insert Table 8 Here]

The results of this test is provided in Table 8 Panel A. The results are still statistically significant for homophily due to native place and native language for all the three regression specifications. Like, for compensation regression, the coefficient of *Homophily_Place* is 0.076 and it is statistically significant at 10 %. Similarly, with the same specification, the coefficient of *Homophily_Language* is 0.142 and it is statistically significant at 5 %. For the Tobin's Q regression, the coefficients are 0.176 (significant at 5 %) and 0.209 (significant at 5 %). The coefficient of *Homophily_Place* and *Homophily_Language* for Residual Q are also positive and significant. The detailed table for this test is provided in Appendix.

6.5.2 Geographical Differences

As maps in Figure 3 shows that the there is a dominance of people from southern part of India at CEO or CEO equivalent positions or at directors' posts, therefore, it is important to check the difference between the homophily arising between a manager and the board members due to identities from southern part of India w.r.t to northern part of India. Also, as southern part of India has different culture, language, food habits, and life style from northern part of India, therefore, there is a possibility that intensity of homophily between a manager and the board members can be different for these two geographical locations. To verify that, I created the dummy variable (*'South'*) which takes value 1 if native place is from southern states - Karnataka, Tamil Nadu, Andhra Pradesh, or Kerala ; otherwise 0. I interact this dummy variable with homophily variables and use the following regression specification:

For the Total Compensation:

$$\begin{aligned} TotalCompensation_{i,t} = & \alpha + \beta_1 South_t + \beta_2 Homophily_{i,t} + \beta_3 HomophilyXSouth_{i,t} + \beta_4 Size_{i,t} + \\ & \beta_5 FirmAge_{i,t} + \beta_6 InstitutionalOwnership_{i,t} + \beta_7 PromotersOwnership_{i,t} + \beta_8 ROA_{i,t} + \gamma_{Designation} + \\ & \gamma_{firm} + \gamma_{year} + \epsilon_t \end{aligned}$$

For the Tobin's Q or Residual Q:

$$\begin{aligned} TobinQ_t(orResidualQ_t) = & \alpha + \beta_1 South + \beta_2 Homophily_t + \beta_3 Homophily_tXSouth + \beta_4 Size + \\ & \beta_5 FirmAge + \beta_6 InstitutionalOwnership + \beta_7 PromotersOwnership + \beta_8 Leverage + \gamma_{firm} + \\ & \gamma_{year} + \epsilon_t \end{aligned}$$

The results for this regressions are provided in Table 8 Panel B. The coefficient of *Homophily_Place* and *Homophily_Language* are positive and significant for all the specifications. This supports the previous results and the main argument of the paper that there is in-group favoritism in deciding the managerial compensation based on native language and native place but because homophily based on these identities also increases the firm value in the long run, therefore, it is beneficial for the firm to bear this cost. Thus, the net benefit from homophily between a manager and the board members shows that the homophily actually reduces the agency frictions.

On the other side, the coefficient of *Homophily X South* is negative but insignificant in all the regressions. It implies that, although managerial positions and directors' posts are dominated by people from southern part of India, the association of homophily with total compensation, long term firm value, and residual firm value does not differ due to homophily based on identities from southern region of India or from northern region of India.

6.5.3 Powerful CEO

As powerful manager has higher influence in decision making process in the organization, therefore, there is a possibility that he can also affect executive compensation decision and firm performance differently. Adams et al.(2005) show that the firm in which CEO has greater power to influence decisions also has higher performance variability. To mitigate this concerns , I also control for manager's influence in the organization. Following Adams et al.(2005), I measure powerful manager (*Powerful_CEO*) as a dummy variable which takes value 1 if either manager is also a founder (promoter) of the firm or manager's concentration of job titles i.e. he is chairman, managing director or CEO of the firm. To test this conjecture, following empirical specifications are used:

For the Total Compensation:

$$TotalCompensation_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Powerful_CEO_{i,t} + \beta_3 Size_{i,t} + \beta_4 FirmAge_{i,t} + \beta_5 InstitutionalOwnership_{i,t} + \beta_6 PromotersOwnership_{i,t} + \beta_7 ROA_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

For the Tobin's Q or Residual Q:

$$TobinQ_{i,t}(or ResidualQ_{i,t}) = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Powerful_CEO_{i,t} + \beta_3 Size_{i,t} + \beta_4 FirmAge_{i,t} + \beta_5 InstitutionalOwnership_{i,t} + \beta_6 PromotersOwnership_{i,t} + \beta_7 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

Results are provided in Table 9 and it shows that even controlling for manager's influence in the organization, homophily due to language and place are positively associated with executive compensation and firm value. This further strengthen the main results of the paper. I also find that there is a positive and statistically significant relation between powerful manager and executive compensation but there is no relation between powerful manager and firm performance. This implies rent extraction due to the manager's influence on decision

making process in the organization. As this is not the main result of the paper, therefore, it needs further research.

6.5.4 College Ties and Past Experience

Farcassi and Tate (2012) show that powerful CEO are more likely to appoint directors with ties to the CEO and that CEO-director ties reduce firm value. CEO-director ties measured in the form of Social Network Index (SNI) and it is defined as the sum of Current Employment Connection, Prior Employment Connection, Education Connection, and Other Activity Connection. From these network connections, The most common are past employment and other activities and the least common are education and current employment. As there is a possibility that increase in compensation, firm value, and residual firm value are not due to the homophily but due to the college ties or past employment network between a manager and the directors and homophily variable is just a proxy for these ties. To mitigate this concern, I also control for college ties and the past employment connections between a manager and the director. Using BoardEx's education database and the network database, I created the variables- *College_Ties* and *Exp_Ties*. *College_Ties* is a dummy variable which takes value 1 if there is at least one director in the board which is from the same college as manager; otherwise 0. Similarly, *Exp_Ties* is a dummy variable which takes value 1 if manager has a past experience of working with at least one director of the board; otherwise 0. The empirical specification to test conjecture is following:

For compensation:

$$TotalCompensation_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 College_Ties(or\ Exp_Ties) + \beta_3 Size_{i,t} + \beta_4 FirmAge_{i,t} + \beta_5 InstitutionalOwnership_{i,t} + \beta_6 PromotersOwnership_{i,t} + \beta_7 ROA_{i,t} + \gamma Designation + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

For firm value and residual value:

$$\begin{aligned} TobinQ_{i,t}(or ResidualQ_{i,t}) = & \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 College_Ties(or Exp_Ties) + \beta_3 PSize_{i,t} + \\ & \beta_4 FirmAge_{i,t} + \beta_5 InstitutionalOwnership_{i,t} + \beta_6 PromotersOwnership_{i,t} + \beta_7 Leverage_{i,t} + \\ & \gamma_{firm} + \gamma_{year} + \epsilon_{i,t} \end{aligned}$$

The results of this test are provided in Table 10. The Panel A shows the results controlling for college ties. The $\beta_{language}$ is 0.370 for Tobin's Q and Residual Q and it is statistically significant. This implies that homophily based on native language between a manager and the board increases firm value and residual firm value even controlling for college ties. Regarding compensation results, although there is a positive association between homophily and total compensation but it is not statistically significant. I also find that college ties between a manager and the board does not impact compensation, firm value, and residual firm value.

Panel B shows the results controlling for ties from past employment. $\beta_{language}$ and β_{place} are positive and statistically significant for firm value and residual firm value. This confirms the main results of the paper that homophily based on native language and native place reduces the agency frictions by increasing the residual firm value. Like in case of college ties, there is a positive association between homophily and total compensation but it is not statistically significant. I also find that ties due to past employment reduces total compensation, firm value, and residual firm value. These results are in line with Farcassi and Tate (2012) findings.

7 Concluding Remarks

This paper examines how the identities affect executive compensation, agency frictions, and the future firm value. I begin with providing an intuition on how identities can affect the agency frictions between a manager and the board by using the Akerlof and Kranton (2005) suggested agent utility function with an group identity (principal and the agent share the same identity) in the simple static model of Holmstrom and Milgrom (1979). This small change in the framework shows that the manager gets higher compensation due to in-group bias, and the same motivates the manager to exert more efforts for the firm and this is reflected in the higher future firm value compare to its counterparts.

To test this empirically, a novel dataset on Indian surnames using the data on 474 million Indians from the Social Economic Caste Census 2011 and the data from Linguistic Survey of India is developed. Indian surnames provide information on native language, native place, caste, and other identities about a individual and therefore, its a single source of multiple identities. Mapping this data with directors and managers surnames provide their identities.

Next, I test whether the group identity (or homophily) between the board and the manager based on above mentioned traits results in any kind of in-group favoritism and if yes, then does the manager reciprocate that favor. I find that the manager gets higher compensation if the manager speaks the same native language or belongs to the same native place as majority of the board members. Furthermore, I also show that such firms performs better than their counterparts. These results are consistent even after exclusion of the cost of in-group favoritism implying that the manager provides extra effort due to the group identity. This result that the individual puts extra effort for their group members compared to outsiders is in line with literature on group identity in social psychology and

social neuroscience.

To check the external validity of the dependent variables, total compensation is replaced with salary and Tobin's Q is replaced with sales growth. Results are robust to these changes. Furthermore, to check the external validity of the homophily constructs, an alternate dataset on surnames with same identities from 'Indianchildnames.com' is used. The results are also robust to this alternate dataset.

As there is a possibility of self-selection that the board can select the manager of the same identity to reduce frictions in corporate decision making, I use independent director's rotation within the same gender as the exogenous shock to the identity of the board. Keeping the replacement of independent director within the same gender maintains the board characteristics like independence and board diversity along with the board size. The results are robust even after tackling the endogeneity concerns.

Finally, I check the main results against the alternative conjectures such as influence of powerful CEO on the firm and the role of college ties or ties from past employment between CEO and board members. The main results robust to the influence of CEO on the firm. Regarding college and ties from past employment, I find that the firm value and the residual value are still positively associated with homophily based on native language and native place even controlling for college ties and ties due to past employment. The coefficient of ties due to past employment suggests that the total managerial compensation, firm value, and residual firm value reduces with manager-board members ties from past employment. The latter results are in line with Farcassi and Tate (2012).

In sum, using the CEO-board setting, this article provides the evidence that how in-group favoritism can result into higher compensation for the manager and how the manager returns this favor by increasing the firm value. This illustrates how group identity can reduce the agency frictions. These results have implications as it shows how homophily based on native language and native place affects the corporate governance, managerial compensation, and

firm value in the emerging economy like India.

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Variable Definition

Variable	Description	Source
TotalCompensation	Log of manager's total compensation. Total compensation is in USD.	ProwessDx
Salary	Log of manager's salary. Salary is in USD.	ProwessDx
TobinQ	Ratio of market value of equity to book value of equity.	ProwessDx
SalesGrowth	Growth in sales compare to last year	ProwessDx
Residual Q	Residual from regression of Tobin's Q on last year managerial compensation	ProwessDx
Homophily_Language	Dummy variable which takes value 1 if a manager and majority of board members have the same native language; otherwise 0.	Linguistic Survey of India (LSI)
Homophily_Place	Dummy variable which takes value 1 if a manager and majority of board members have the same native place; otherwise 0.	Socio-Economic Caste Census (SECC)
Homophily_Caste	Dummy variable which takes value 1 if a manager and majority of board members have the same caste; otherwise 0.	Socio-Economic Caste Census (SECC)
Size	Log of market capitalization	ProwessDx
ROA	Ratio of net income to total assets.	ProwessDx
PromotersOwnership	Perentage of equity ownership by promoters.	ProwessDx
InstitutionalOwnership	Perentage of equity ownership by institutional investors.	ProwessDx
Leverage	Ratio of total debt to total assets.	ProwessDx
South	Dummy variable which takes value 1 if the state is in south part of India; otherwise 0.	Socio-Economic Caste Census (SECC)
Firm Age	Age of the firm from incorporation	ProwessDx
Board Size	Number of board members	BoardEx
Board Independence	Percentage of independent board members	BoardEx
Board Diversity	Percentage of women in board	BoardEx
ID Rotation	Dummy variable which takes value 1 if there is change in independent director in the same gender	BoardEx

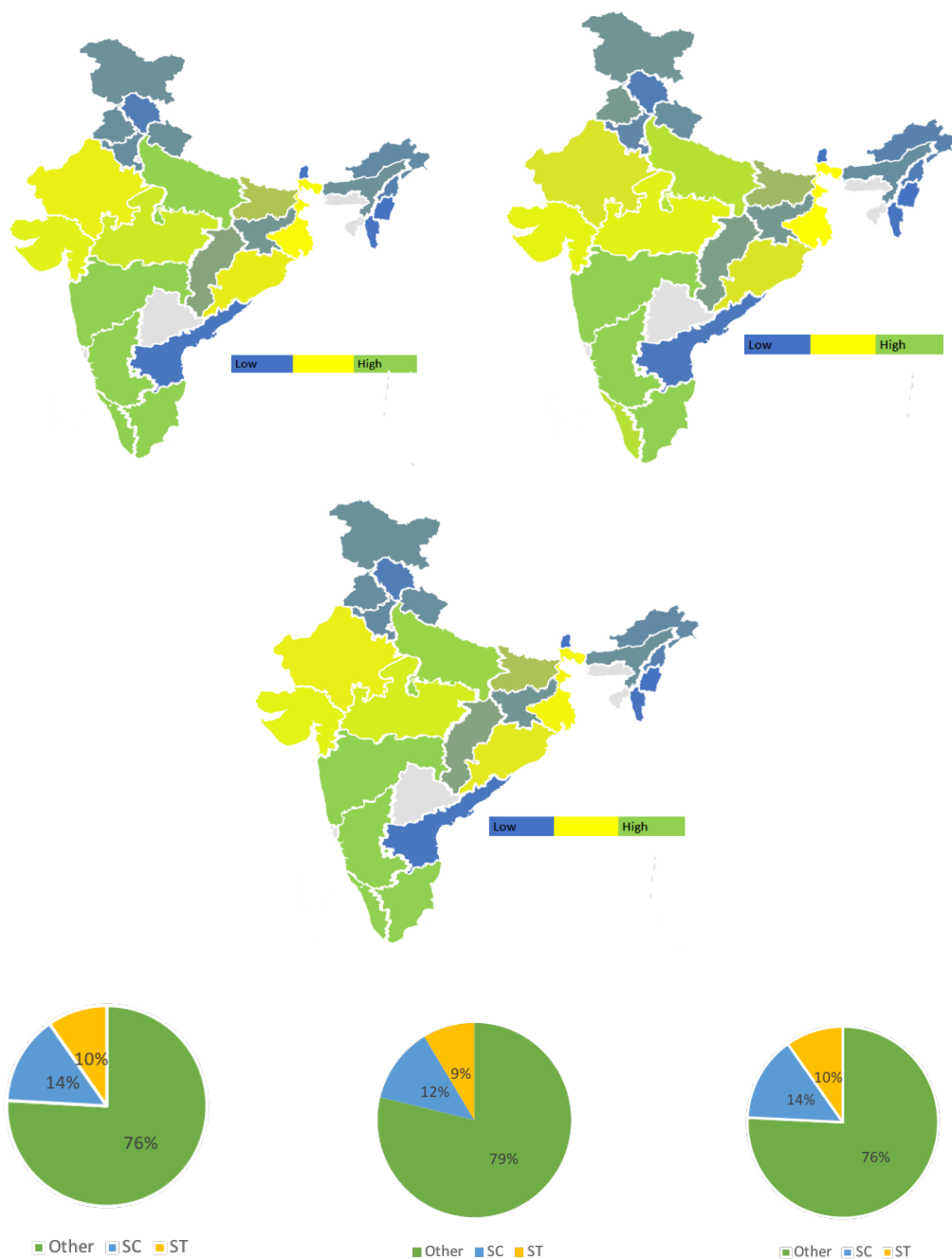


Figure 1: Maps showing the distribution of place of origin among all Indian directors, managers, and the remaining directors. The pie charts show the distribution of caste among all Indian directors, managers, and the remaining directors.

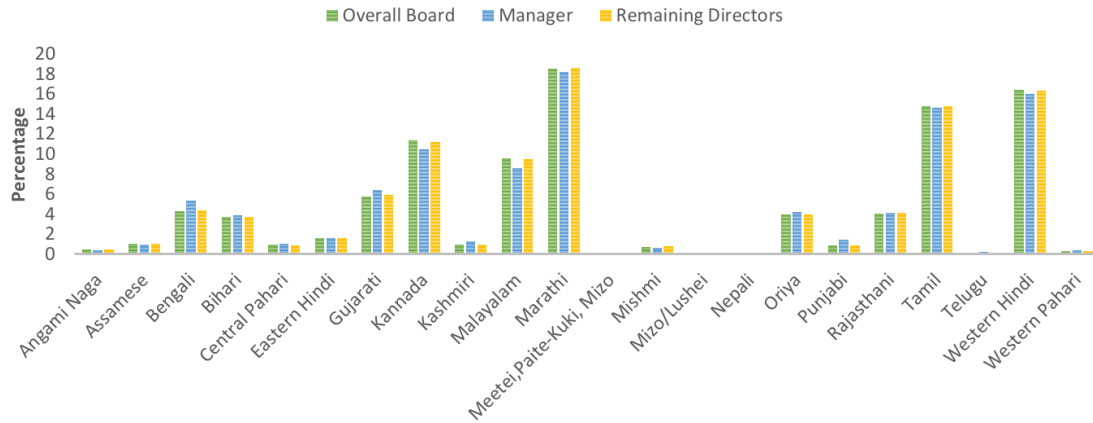


Figure 2: This histogram shows the distribution of language among all Indian directors, managers, and the remaining directors.

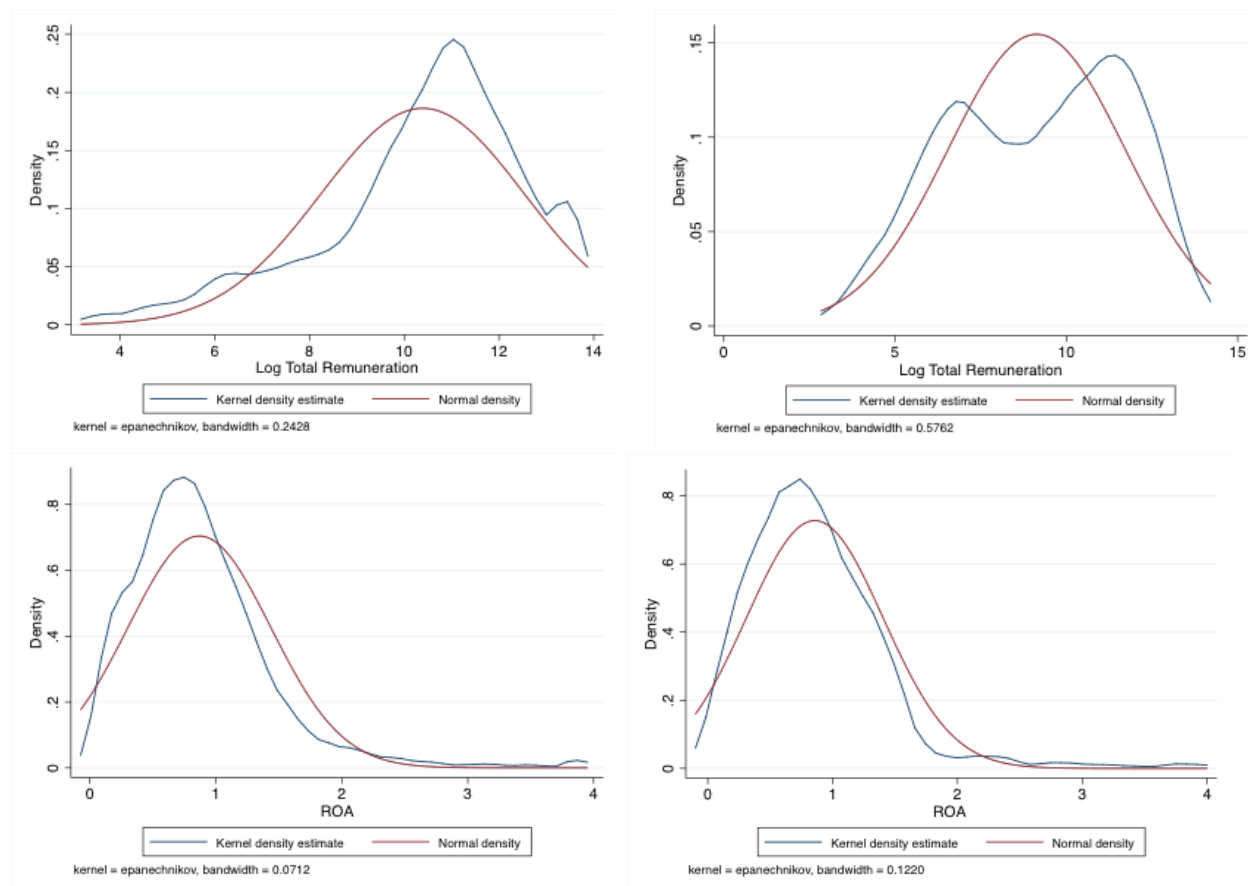


Figure 3: Compensation and Firm Performance : Promoter versus Non-Promoter

Table 1
Panel A: Summary Statistics

This table provides the summary statistics of the main variables. Log Remuneration is log of total compensation. Log Salary is log of salary. Tobin Q is ratio of market price of equity to the book value of equity. Sales Growth is growth of the sales w.r.t. last year. Residual Q is residual of regression of Tobin Q on manager's total compensation of last year. Homophily Language is a dummy variable whose value equal to 1 if the board (on average) and the manager speaks same Indian language otherwise 0. Homophily Place is dummy variable whose value equal to 1 if the board and the manager belong to same Indian state otherwise 0. Homophily Caste is dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Leverage is total debt by total assets. Business Group is dummy variable whose value equal to 1 if firm belongs to any business group otherwise 0. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. Independent Board is percentage of independent directors to the total number of directors. Gender Diversity is percentage of women in board. Number of Directors is total number of directors in board.

Variable	Observations	Mean	Std. Dev.	Min	Max
Dependent					
Log Remuneration	41,499	10.239	2.174	3.412	13.634
Log Salary	32,563	10.700	1.357	6.016	13.493
Tobin Q	30,249	2.231	3.066	-3.070	19.260
Sales Growth	16,627	0.145	0.568	-0.999	7.067
Residual Q	30,085	-0.019	1.749	-5.143	8.470
Independent					
Homophily Language	41,507	0.459	0.498	0.000	1.000
Homophily Place	41,507	0.440	0.496	0.000	1.000
Homophily Caste	41,507	0.898	0.303	0.000	1.000
Controls					
Size (log Market Cap)	39,438	16.982	1.540	14.103	21.463
Firm Age	41,493	31.583	19.659	4.000	102.000
Leverage	17,364	0.325	0.231	0.001	1.692
Business Group	41,507	0.418	0.493	0.000	1.000
Promoters Ownership	39,438	53.337	17.192	5.960	88.580
Promoter - Manager	15,354	0.931	0.254	0.000	1.000
Institutional Ownership	39,438	8.014	11.305	0.000	48.365
Independent Board (in %)	4,451	50.214	11.176	0.000	80.000
Gender Diversity	4,451	0.098	0.079	0.000	0.429
Number of Directors	4,451	11.021	4.949	4.000	34.000

Panel B: Homophily Distribution (For Managers)

Homophily Language is dummy variable whose value equal to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is dummy variable whose value equal to 1 if the board and the manager belong to same Indian state otherwise 0. Homophily Caste is dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0.

	Language		Place		Caste	
Homophily	Freq.	Percent	Freq.	Percent	Freq.	Percent
0	22,439	54.06	23,226	55.96	4,244	10.22
1	19,068	45.94	18,281	44.04	37,263	89.78
Total	41,507	100	41,507	100	41,507	100

Panel C: Managerial Compensation - Distribution and Summary Statistics

This table shows the components of managerial compensation and their summary statistics. Total Remuneration is total managerial compensation in USD. Salary is major component of total managerial compensation and it is in USD. Variable pay comprises Retirement Benefits and Perks. Both are in USD.

Variable	Obs	Mean	Std. Dev.	Min	Max
Total Remuneration	41,507	144179.8	302414.2	74.6025	2024083
Salary	32,567	101644.1	156317.1	1333.645	965457.6
Retirement Benefits	2,405	27964.32	49362.24	148.5924	314630
Perks	12,735	34156.37	67381.49	79.4865	456702.4

Table 2
Group Identity and Managerial Compensation (Baseline Scenario)

Total compensation is log of total compensation paid to the manager. Here manager is equal to CEO or CEO equivalent positions. CEO equivalent positions are chairman, president, and managing director. Homophily Language is a dummy variable whose value equals to 1 if the board (on average) and the manager speaks same Indian language otherwise 0. Homophily Place is a dummy variable whose value equals to 1 if the board and the manager belong to same Indian state otherwise 0. Homophily Caste is dummy variable whose value equals to 1 if the board (on average) and the manager belong to same caste otherwise 0. The empirical specification is :

$$TotalCompensation_{i,t} = \alpha + \beta Homophily_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year}$$

$\gamma_{Designation}$, γ_{firm} , and γ_{year} are fixed effects for managerial designation, firm, and for year. Homophily is based on language, place, or caste.

VARIABLES	(1) Total Compensation	(2) Total Compensation	(3) Total Compensation
Homophily Language	0.052*** (0.017)		
Homophily Place		0.035** (0.017)	
Homophily Caste			-0.072** (0.032)
Constant	10.224*** (0.010)	10.233*** (0.009)	10.312*** (0.030)
Observations	41,184	41,184	41,184
R-squared	0.767	0.767	0.767
Managerial Designation	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Table 3
Group Identity and Managerial Compensation (With Controls)

Total compensation is log of total compensation paid to the manager. Here manager is equal to CEO or CEO equivalent positions (i.e. chairman, president, and managing director). Homophily Language is dummy variable whose value equal to 1 if the board (on average) and the manager speaks same Indian language otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. ROA is return on assets calculated as ratio of net income to total assets. The empirical specification is:

$$TotalCompensation_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 ROA_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year}$$

$\gamma_{Designation}$, γ_{firm} , and γ_{year} are fixed effects for managerial designation, firm, and for year. Homophily is based on language, place, or caste.

	(1)	(2)	(3)
VARIABLES	Total Compensation	Total Compensation	Total Compensation
Homophily Language	0.140*** (0.041)		
Homophily Place		0.077* (0.042)	
Homophily Caste			-0.016 (0.077)
Observations	17,626	17,626	17,626
R-squared	0.483	0.483	0.483
Managerial Designation	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Table 4
Group Identity and Firm Performance

This table shows the regression results of firm value (measured as Tobin Q) on group identity(measured as Homophily). Tobin Q is the ratio of market price of a equity to the book value of a equity. Homophily Language is a dummy variable whose value equal to 1 if the board (on average) and the manager speaks same Indian language otherwise 0. Homophily Place is dummy variable whose value equal to 1 if the board and the manager belong to the same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. ROA is return on assets calculated as ratio of net income to total assets. The empirical specification is :

$$TobinQ_{i,t} = \alpha + \beta_{1,t}Homophily_{i,t} + \beta_2Size_{i,t} + \beta_3FirmAge_{i,t} + \beta_4InstitutionalOwnership_{i,t} + \beta_5PromotersOwnership_{i,t} + \beta_6Leverage_{i,t} + \gamma_{firm} + \gamma_{year}$$

γ_{firm} and γ_{year} are fixed effects for firm and for year. Homophily is based on language, place, or caste.

	(1)	(2)	(3)
VARIABLES	Tobin Q	Tobin Q	Tobin Q
Homophily Language	0.205*** (0.055)		
Homophily Place		0.168*** (0.057)	
Homophily Caste			-0.164 (0.108)
Observations	15,172	15,172	15,172
R-squared	0.590	0.589	0.589
Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Table 5
Group Identity and Residual Q

This table shows the regression results of residual firm value (measured as Residual Q) on group identity(measured as Homophily). Residual Q is the residual of regressing TobinQ on last year's managerial total compensation. Homophily Language is a dummy variable whose value equal to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. Leverage is the ratio of total debt by total assets. The empirical specification is :

$$ResidualQ_{i,t} = \alpha + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

γ_{firm} and γ_{year} are fixed effects for firm and for year. Homophily is based on language, place, or caste.

VARIABLES	(1) Residual Q	(2) Residual Q	(3) Residual Q
Homophily Language	0.148*** (0.047)		
Homophily Place		0.139*** (0.049)	
Homophily Caste			-0.147 (0.092)
Observations	15,165	15,165	15,165
R-squared	0.129	0.129	0.129
Control	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Table 6
Panel A: External Validity of Dependent Variable

This table shows the regression results using the alternative dependent variables. Salary is log of salary. Sales Growth is sales growth relative to last year. Homophily Language is a dummy variable whose value equals to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. ROA is return on assets calculated as ratio of net income to total assets. The empirical specification for these tests are:

$$Salary_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 ROA_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

$$Sales_Growth_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

VARIABLES	(1) Salary	(2) Salary	(3) Salary	(4) Sales Growth	(5) Sales Growth	(6) Sales Growth
Homophily Language	0.031* (0.019)			0.031*** (0.009)		
Homophily Place		0.057*** (0.019)			0.028*** (0.009)	
Homophily Caste			-0.128*** (0.030)			0.009 (0.014)
Observations	12,990	12,990	12,990	15,612	15,612	15,612
R-squared	0.256	0.256	0.257	0.076	0.076	0.076
Managerial Designation	Yes	Yes	Yes	No	No	No
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes

Panel B : External Validity Test of Homophily Construct

This table shows the regression results of using alternative homophily variable. To check the external validity of the homophily variables, rather than using the SECC 2011 and LSI data, I use the data from Indianchildnames.com. The empirical specification for this test is,

$$TotalCompensation_{i,t} = \alpha_i + \beta_1 In - Group_{Favoritism_{i,t}} + \beta_2 Size_{i,t} + \beta_3 ROA_{i,t} + \beta_4 Leverage_{i,t} + \beta_5 Board_Size_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

Here *TotalCompensation* means either total compensation or the salary and *In-Group-Favoritism* is an group identity dummy based on native language. Size is log of market capitalization. ROA is return on assets calculated as ratio of net income to total assets. Leverage is the ratio of total debt by total assets. Board Size is total number of board members.

VARIABLES	(1) Total Compensation	(2) Salary
In-Group Favoritism	0.138*** (0.032)	0.102*** (0.032)
Size	0.368*** (0.029)	0.337*** (0.024)
ROA	1.364*** (0.136)	0.854*** (0.145)
Leverage	-0.308*** (0.091)	-0.228** (0.089)
Board Size	0.004 (0.007)	-0.002 (0.005)
Constant	7.935*** (0.226)	8.016*** (0.189)
Observations	14,523	14,229
R-squared	0.805	0.804
Firm Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes

Table 7
Endogeneity Issue and the Identification Test

This table shows the regression results for identification test. Total Compensation is log of total compensation. TobinQ is the ratio of market price of an equity to the book value of an equity. Residual Q is the TobinQ condition on managerial compensation of last year. Homophily Language is a dummy variable whose value equal to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. ID Rotation is dummy variable which takes value 1 if the independent board member replaced by another independent board member within the same gender ; otherwise 0. The empirical specification for this test is,

$$Y_{i,t} = \alpha_i + \beta_1 IDRotationXHomophily_{i,t} + \beta X_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

Here Y is total compensation (*Total Compensation*), firm performance (*TobinQ*), or firm performance conditioned on managerial compensation (*ResidualQ*).

[illegible]

Table 8
Panel A: Community Migration and Group Identity

This table shows the regression results excluding the surnames for those communities which migrated from part of India to another after independence. Total Compensation is log of total compensation. TobinQ is the ratio of market price of an equity to the book value of an equity. Residual Q is the TobinQ condition on managerial compensation of last year. Homophily Language is a dummy variable whose value equals to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. ROA is return on assets calculated as ratio of net income to total assets. The empirical specification for these tests are:

$$TotalCompensation_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 ROA_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

$$TobinQ_{i,t}(or ResidualQ_{i,t}) = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Size_{i,t} + \beta_3 FirmAge_{i,t} + \beta_4 InstitutionalOwnership_{i,t} + \beta_5 PromotersOwnership_{i,t} + \beta_6 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

VARIABLES	(1) Total_Compensation	(2) Total_Compensation	(3) Total_Compensation	(4) Tobin's Q	(5) Tobin's Q	(6) Tobin's Q	(7) Residual Q	(8) Residual Q	(9) Residual Q
Homophily Place	0.076* (0.042)			0.176*** (0.057)			0.144*** (0.042)		
Homophily Language		0.142*** (0.041)			0.209*** (0.055)			0.150*** (0.047)	
Homophily Caste			-0.018 (0.079)			-0.101 (0.109)			-0.084 (0.093)
Constant	7.217*** (1.239)	7.153*** (1.239)	7.272*** (1.241)	-9.758*** (1.723)	-9.755*** (1.723)	-9.547*** (1.726)	-10.405*** (1.468)	-10.419*** (1.468)	-10.245*** (1.471)
Observations	17,371	17,371	17,371	14,950	14,950	14,950	14,943	14,943	14,943
R-squared	0.485	0.486	0.485	0.583	0.583	0.583	0.131	0.131	0.131
Manager Designation	Yes	Yes	Yes	No	No	No	No	No	No
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: Geographical Differences

This table shows the regression results excluding the surnames for those communities which migrated from part of India to another after independence. South is dummy variable which takes value if a manager comes from southern Indian states; otherwise 0. Total Compensation is log of total compensation. TobinQ is the ratio of market price of an equity to the book value of an equity. Residual Q is the TobinQ condition on managerial compensation of last year. Homophily Language is a dummy variable whose value equals to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. ROA is return on assets calculated as ratio of net income to total assets. The empirical specification for these tests are:

$$TotalCompensation_{i,t} = \alpha + \beta_1 South + \beta_2 Homophily_{i,t} + \beta_3 HomophilyxSouth_{i,t} + \beta_4 Size_{i,t} + \beta_5 FirmAge_{i,t} + \beta_6 InstitutionalOwnership_{i,t} + \beta_7 PromotersOwnership_{i,t} + \beta_8 ROA_{i,t} + \gamma_{Designation} + \gamma_{firm} + \gamma_{year} + \epsilon_t$$

$$TobinQ_{i,t}(or ResidualQ_{i,t}) = \alpha_i + \beta_1 South + \beta_2 Homophily_{i,t} + \beta_3 HomophilyxSouth_{i,t} + \beta_4 Size_{i,t} + \beta_5 FirmAge_{i,t} + \beta_6 InstitutionalOwnership_{i,t} + \beta_7 PromotersOwnership_{i,t} + \beta_8 Leverage_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

VARIABLES	(1) Total_Compensation	(2) Total_Compensation	(3) Total_Compensation	(4) Tobin's Q	(5) Tobin's Q	(6) Tobin's Q	(7) Residual Q	(8) Residual Q	(9) Residual Q
South	0.091 (0.061)	0.078 (0.062)	0.078 (0.226)	-0.109 (0.082)	-0.110 (0.083)	-0.360 (0.328)	-0.115* (0.070)	-0.119* (0.071)	-0.372 (0.280)
Homophily Place	0.105** (0.047)			0.188*** (0.064)			0.159*** (0.054)		
Homophily Place x South	-0.120 (0.090)			-0.090 (0.123)			-0.090 (0.105)		
Homophily Language		0.152*** (0.045)			0.220*** (0.061)			0.161*** (0.052)	
Homophily Language x South		-0.054 (0.089)			-0.068 (0.121)			-0.062 (0.103)	
Homophily Caste			-0.020 (0.082)			-0.178 (0.113)			-0.161* (0.096)
Homophily Caste x South			-0.012 (0.225)			0.244 (0.327)			0.248 (0.279)
Constant	7.298*** (1.237)	7.229*** (1.237)	7.343*** (1.239)	-9.567*** (1.721)	-9.571*** (1.720)	-9.294*** (1.724)	-10.332*** (1.469)	-10.323*** (1.469)	-10.093*** (1.471)
Observations	17,626	17,626	17,626	15,172	15,172	15,172	15,165	15,165	15,165
R-squared	0.483	0.484	0.483	0.590	0.590	0.589	0.130	0.130	0.129
Manager Designation	Yes	Yes	Yes	No	No	No	No	No	No
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 9
Powerful CEO

This table shows the regression results including the variable on CEO power. Total Compensation is log of total compensation. TobinQ is the ratio of market price of an equity to the book value of an equity. Residual Q is the TobinQ condition on managerial compensation of last year. Homophily Language is a dummy variable whose value equals to 1 if the board (on average) and the manager speaks same Indian language, otherwise 0. Homophily Place is a dummy variable whose value equal to 1 if the board and the manager belong to same Indian state, otherwise 0. Homophily Caste is a dummy variable whose value equal to 1 if the board (on average) and the manager belong to same caste, otherwise 0. Size is log of market capitalization. Powerful_CEO is a dummy variable which takes value 1 if CEO Firm Age is firm age from year of incorporation. Promoters Ownership is percentage of promoters equity holding to the total equity issued by the firm. Institutional Ownership is percentage of institutional investors equity holding to the total equity issued by the firm. ROA is return on assets calculated as ratio of net income to total assets. The empirical specification for these tests are:

$$TotalCompensation_{i,t} = \alpha_i + \beta_1 Homophily_{i,t} + \beta_2 Powerful_CEO_{i,t} + \beta_3 Size_{i,t} + \beta_4 FirmAge_{i,t} + \beta_5 InstitutionalOwnership_{i,t} + \beta_6 PromotersOwnership_{i,t} + \beta_7 ROA_{i,t} + \gamma_{firm} + \gamma_{year} + \epsilon_{i,t}$$

$$\begin{aligned} \text{Tobin}Q_{i,t}(\text{or Residual}Q_{i,t}) &= \alpha_i + \beta_1 \text{Homophily}_{i,t} + \beta_2 \text{Powerful_CEO}_{i,t} + \beta_3 \text{Size}_{i,t} + \beta_4 \text{FirmAge}_{i,t} + \\ &\beta_5 \text{InstitutionalOwnership}_{i,t} + \beta_6 \text{PromotersOwnership}_{i,t} + \beta_7 \text{Leverage}_{i,t} + \gamma_{\text{firm}} + \gamma_{\text{year}} + \epsilon_{i,t} \end{aligned}$$

[illegible]

Appendix

Appendix A

A1: Data Filtering

	Firms	Observations
Merging with surname and social identities data (Dataset-I)	24326	1473836
Total remuneration is given (Dataset-II)	13226	285159
BSE and NSE listed non- financial firms (Overall Board)	3668	208275
BSE and NSE listed non- financial firms (Managers)	3668	41507

A2: Yearly Distribution

Year	Freq.	Percent	Cum.
2004	2,011	4.84	4.84
2005	2,177	5.24	10.09
2006	2,410	5.81	15.9
2007	2,551	6.15	22.04
2008	2,608	6.28	28.33
2009	2,670	6.43	34.76
2010	2,739	6.6	41.36
2011	2,748	6.62	47.98
2012	2,822	6.8	54.78
2013	2,797	6.74	61.51
2014	2,713	6.54	68.05
2015	2,972	7.16	75.21
2016	3,512	8.46	83.67
2017	3,607	8.69	92.36
2018	3,170	7.64	100
Total	41,507	100	

A3: Business Group Distribution

Business Group	Freq.	Percent	Cum.
0	24,156	58.2	58.2
1	17,351	41.8	100
Total	41,507	100	

Appendix B

B1: Distribution of Manager Designation

Designation	Freq.	Percent	Cum.
Chairperson	9627	23.27	23.27
Chairperson & Chief Executive Officer	137	0.33	23.6
Chairperson & Chief Mentor	7	0.02	23.62
Chairperson & Co. Secretary	7	0.02	23.64
Chairperson & Director	356	0.86	24.5
Chairperson & Executive Director	778	1.88	26.38
Chairperson & Joint Managing Director	33	0.08	26.46
Chairperson & Managing Director	9897	23.93	50.38
Chairperson & President	16	0.04	50.42
Chairperson Emeritus	139	0.34	50.76
Chairperson, Managing Director & Chief Executive Officer	184	0.44	51.2
Chief Executive Officer	720	1.74	52.94
Chief Executive Officer & Deputy Managing Director	5	0.01	52.96
Chief Executive Officer & Director	26	0.06	53.02
Chief Executive Officer & Executive Director	93	0.22	53.24
Chief Executive Officer & Manager	5	0.01	53.26
Chief Executive Officer & President	40	0.1	53.35
Director & Chief Executive Officer	136	0.33	53.68
Executive Director & Chief Executive Officer	571	1.38	55.06
Managing Director	15700	37.95	93.02
Managing Director & Chief Executive Officer	1299	3.14	96.16
Managing Director & Chief Operations Officer	13	0.03	96.19
Managing Director & Co. Secretary	23	0.06	96.24
Managing Director & President	60	0.15	96.39
President	190	0.46	96.85
President & Chief Financial Officer	38	0.09	96.94
President & Director	46	0.11	97.05
Vice Chairperson & Managing Director	1220	2.95	100
Total	41507	100	

Appendix C

The SECC 2011 data contains information on individual name, parents name, gender, age, caste, address (state and district) and others. This information on 474 million Indians is available in 48 CSV files. The surname data developed from SECC provide information on surname, state, district, and caste. To scrape surnames from SECC data, the cloud computing on the Microsoft Azure is used with an instance of 32 GB RAM and 1 TB hard disk.

The steps to scrape surnames from Indian names are: firstly, the observations with missing names are removed. Second, all the special characters and names in the native language are replaced by white space. Third, the slicing of the names is done till all surnames are obtained. Fourth, if the surname length is 2 or 1, it means either names are from south India or names are long. In SECC data, south Indian names and long names are recorded differently. For example: names like "P.K. Chandrashekhhar Rao" is penned as "Chandrashekhhar Rao P.K." in the SECC data. Therefore, when surnames are either 1 or 2 length, it means those are not surnames but initials. These initials are replaced by second name if the name has three words such as "Rao" from "Chandrashekhhar Rao P.K." or with first name if a name has two words such as "Gupta" from "Gupta S.K." Fifth, even after all these steps, if there are surnames with length 1 or 2, those surnames are removed to curb noise in the final dataset. Otherwise, final dataset would contain certain discrepancies such as surname "r" which is part of "Sahil Kuma r", but, in this name, surname is "Kumar" and not "r". All the designations such as Shri, Sh., Mr., Mr, Dr., Dr, Col., Maj., and others are removed in the beginning. Also, as Indian names are written differently in northern India and southern India. To mitigate these concerns, various books and websites are referred.

Appendix D

Group Identity and Managerial Compensation (with controls)

VARIABLES	(1) Total Compensation	(2) Total Compensation	(3) Total Compensation
Homophily Language	0.140*** (0.041)		
Homophily Place		0.077* (0.042)	
Homophily Caste			-0.016 (0.077)
Size	0.070** (0.031)	0.070** (0.031)	0.069** (0.031)
Firm Age	0.043 (0.032)	0.043 (0.032)	0.043 (0.032)
Institutional Ownership	0.021*** (0.003)	0.021*** (0.003)	0.020*** (0.003)
Promoters Ownership	0.006*** (0.002)	0.006*** (0.002)	0.006*** (0.002)
ROA	0.307*** (0.050)	0.305*** (0.050)	0.303*** (0.050)
Constant	7.249*** (1.237)	7.312*** (1.237)	7.362*** (1.239)
Observations	17,626	17,626	17,626
R-squared	0.483	0.483	0.483
Managerial Designation	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Group Identity and Tobin's Q

VARIABLES	(1) Tobin Q	(2) Tobin Q	(3) Tobin Q
Homophily Language	0.205*** (0.055)		
Homophily Place		0.168*** (0.057)	
Homophily Caste			-0.164 (0.108)
Size	0.314*** (0.042)	0.317*** (0.042)	0.314*** (0.042)
Firm Age	0.075* (0.044)	0.075* (0.044)	0.075* (0.044)
Institutional Ownership	0.048*** (0.003)	0.048*** (0.003)	0.048*** (0.003)
Promoters Ownership	0.064*** (0.003)	0.064*** (0.003)	0.063*** (0.003)
Leverage	-1.243*** (0.150)	-1.222*** (0.150)	-1.221*** (0.150)
Constant	-9.612*** (1.720)	-9.607*** (1.721)	-9.353*** (1.723)
Observations	15,172	15,172	15,172
R-squared	0.590	0.589	0.589
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Group Identity and Residual Q

VARIABLES	(1) Residual Q	(2) Residual Q	(3) Residual Q
Homophily Language	0.148*** (0.047)		
Homophily Place		0.139*** (0.049)	
Homophily Caste			-0.147 (0.092)
Size	0.296*** (0.036)	0.298*** (0.036)	0.296*** (0.036)
Firm Age	0.049 (0.037)	0.048 (0.037)	0.049 (0.037)
Institutional Ownership	0.046*** (0.003)	0.046*** (0.003)	0.046*** (0.003)
Promoters Ownsership	0.058*** (0.003)	0.057*** (0.003)	0.057*** (0.003)
Leverage	-1.007*** (0.128)	-0.992*** (0.128)	-0.991*** (0.128)
Constant	-10.367*** (1.469)	-10.374*** (1.469)	-10.154*** (1.471)
Observations	15,165	15,165	15,165
R-squared	0.129	0.129	0.129
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Group Identity, Managerial Compensation, and Board Characteristics

VARIABLES	(1) Total Compensation	(2) Total Compensation	(3) Total Compensation
Homophily Language	0.105* (0.055)		
Homophily Place		0.129** (0.057)	
Homophily Caste			-0.227*** (0.087)
Size	0.010 (0.021)	0.012 (0.021)	0.008 (0.021)
Firm Age	0.006*** (0.001)	0.006*** (0.001)	0.005*** (0.001)
Institutional Ownership	0.030*** (0.003)	0.030*** (0.003)	0.030*** (0.003)
Promoters Ownership	0.009*** (0.002)	0.009*** (0.002)	0.009*** (0.002)
ROA	0.265*** (0.050)	0.268*** (0.050)	0.263*** (0.050)
Board Independence	0.026*** (0.002)	0.026*** (0.002)	0.026*** (0.002)
Gender Diversity	1.724*** (0.383)	1.669*** (0.382)	1.680*** (0.381)
Board Size	0.020*** (0.005)	0.020*** (0.005)	0.021*** (0.005)
Constant	8.186*** (0.439)	8.141*** (0.441)	8.482*** (0.440)
Observations	3,667	3,667	3,667
R-squared	0.430	0.431	0.431
Managerial Designation	Yes	Yes	Yes
Year	Yes	Yes	Yes

External Validity - Alternative Dependent Variables

VARIABLES	(1) Salary	(2) Salary	(3) Salary	(4) Sales Growth	(5) Sales Growth	(6) Sales Growth
Homophily Language	0.031* (0.019)			0.031*** (0.009)		
Homophily Place		0.057*** (0.019)			0.028*** (0.009)	
Homophily Caste			-0.128*** (0.030)			0.009 (0.014)
Size	0.119*** (0.007)	0.120*** (0.007)	0.120*** (0.007)	0.003 (0.003)	0.003 (0.003)	0.002 (0.003)
Leverage	-0.020 (0.041)	-0.019 (0.041)	-0.019 (0.041)	-0.107*** (0.019)	-0.106*** (0.019)	-0.106*** (0.019)
Firm Age	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
ROA	0.048*** (0.016)	0.047*** (0.016)	0.049*** (0.016)			
Promoters Ownership	0.007*** (0.001)	0.007*** (0.001)	0.007*** (0.001)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Instituional Ownership	0.032*** (0.001)	0.032*** (0.001)	0.032*** (0.001)	0.001** (0.000)	0.001** (0.000)	0.001** (0.000)
Constant	8.400*** (0.124)	8.379*** (0.124)	8.530*** (0.126)	0.117** (0.058)	0.119** (0.058)	0.131** (0.059)
Observations	12,990	12,990	12,990	15,612	15,612	15,612
R-squared	0.256	0.256	0.257	0.076	0.076	0.076
Managerial Designation	Yes	Yes	Yes	No	No	No
Year	Yes	Yes	Yes	Yes	Yes	Yes

Geographical Differences

VARIABLES	(1) Total_Compensation	(2) Total_Compensation	(3) Total_Compensation	(4) Tobin's Q	(5) Tobin's Q	(6) Tobin's Q	(7) Residual Q	(8) Residual Q	(9) Residual Q
South	0.091 (0.061)	0.078 (0.062)	0.078 (0.226)	-0.109 (0.082)	-0.110 (0.083)	-0.360 (0.328)	-0.115* (0.070)	-0.119* (0.071)	-0.372 (0.280)
Homophily Place	0.105** (0.047)			0.188*** (0.064)			0.159*** (0.054)		
Homophily Place x South	-0.120 (0.090)			-0.090 (0.123)			-0.090 (0.105)		
Homophily Language		0.152*** (0.045)			0.220*** (0.061)			0.161*** (0.052)	
Homophily Language x South		-0.054 (0.089)			-0.068 (0.121)			-0.062 (0.103)	
Homophily Caste			-0.020 (0.082)			-0.178 (0.113)			-0.161* (0.096)
Homophily Caste x South			-0.012 (0.225)			0.244 (0.327)			0.248 (0.279)
Size	0.070** (0.031)	0.069** (0.031)	0.068** (0.031)	0.317*** (0.042)	0.314*** (0.042)	0.314*** (0.042)	0.299*** (0.036)	0.296*** (0.036)	0.296*** (0.036)
Firm Age	0.043 (0.032)	0.044 (0.032)	0.044 (0.032)	0.074* (0.044)	0.075* (0.044)	0.074* (0.044)	0.047 (0.037)	0.048 (0.037)	0.048 (0.037)
Inst Ownership	0.021*** (0.003)	0.021*** (0.003)	0.020*** (0.003)	0.048*** (0.003)	0.048*** (0.003)	0.048*** (0.003)	0.046*** (0.003)	0.046*** (0.003)	0.046*** (0.003)
Promoters Ownership	0.006*** (0.002)	0.006*** (0.002)	0.006*** (0.002)	0.064*** (0.003)	0.064*** (0.003)	0.064*** (0.003)	0.057*** (0.003)	0.058*** (0.003)	0.057*** (0.003)
ROA	0.305*** (0.050)	0.307*** (0.050)	0.304*** (0.050)						
Leverage				-1.216*** (0.150)	-1.238*** (0.150)	-1.216*** (0.150)	-0.986*** (0.128)	-1.003*** (0.128)	-0.985*** (0.128)
Constant	7.298*** (1.237)	7.229*** (1.237)	7.343*** (1.239)	-9.567*** (1.721)	-9.571*** (1.720)	-9.294*** (1.724)	-10.332*** (1.469)	-10.323*** (1.469)	-10.093*** (1.471)
Observations	17,626	17,626	17,626	15,172	15,172	15,172	15,165	15,165	15,165
R-squared	0.483	0.484	0.483	0.590	0.590	0.589	0.130	0.130	0.129
Manager Designation	Yes	Yes	Yes	No	No	No	No	No	No
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes